Advanced Number Theory

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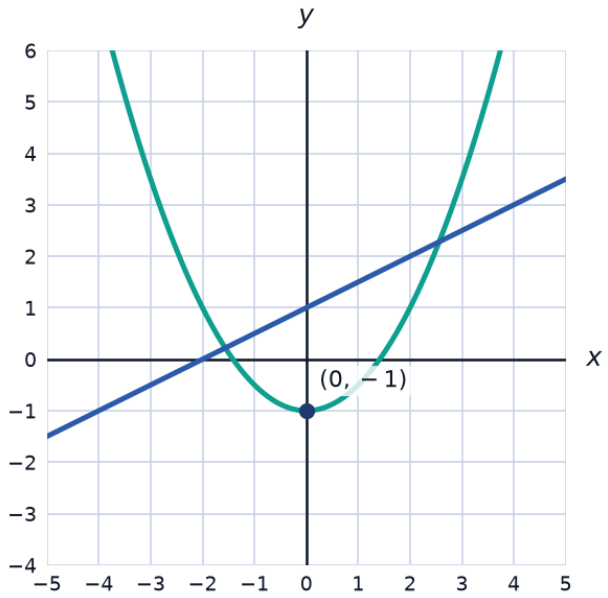
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Divisibility rules and prime factorization

Find the prime factorization of 1260.

2 [Foundational] • Target: Divisibility rules and prime factorization

Determine whether 143 is prime, and justify your answer.



3 [Foundational] • Target: GCF, LCM, and their relationship

Find the GCF and LCM of 36 and 60.

4 [Foundational] • Target: GCF, LCM, and their relationship

Given that two numbers have GCF 8 and LCM 96, and one of the numbers is 24, find the other number, using the identity GCF × LCM = product of the two numbers.

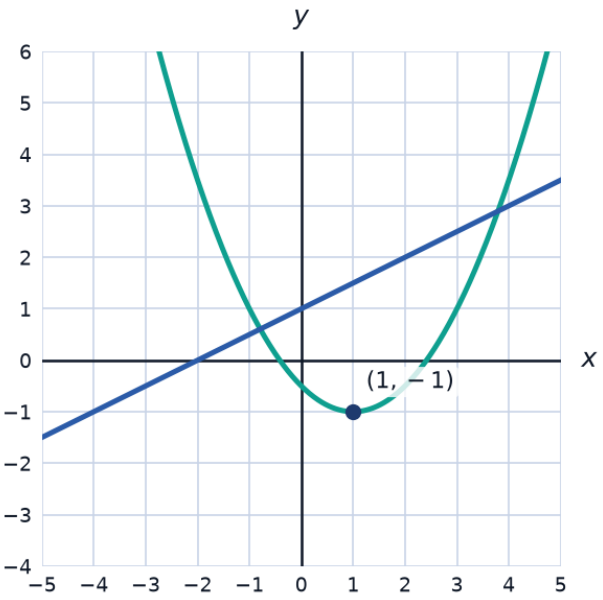
5 [Foundational] • Target: Modular arithmetic

Find the remainder when 47 is divided by 6, and express this using modular notation.

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6 [Foundational] • Target: Modular arithmetic

Find the last digit of 7^{100}, using the pattern of the last digits of powers of 7 (which cycle every 4: 7, 9, 3, 1).



7 [Foundational] • Target: Sums and products of integer sets

Find the sum of all integers from 1 to 100, using the formula $S = \frac{n(n+1)}{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sums and products of integer sets

Find the sum of all even integers from 2 to 200.

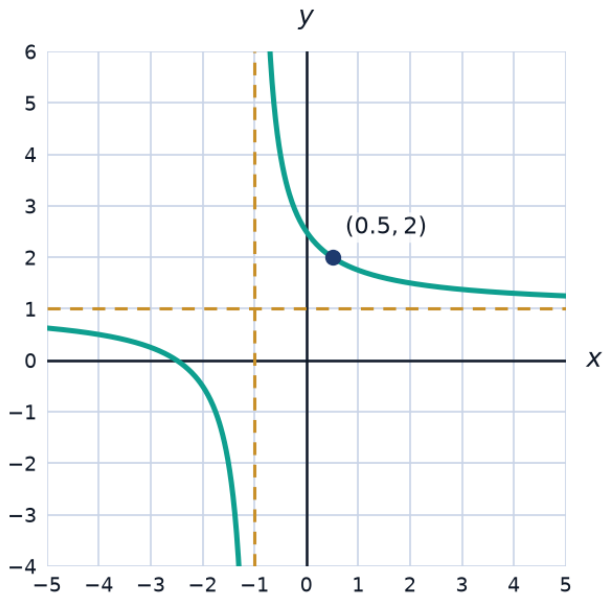
9 [Intermediate] • Target: Number sets and properties

Classify $\sqrt{16}$ as rational or irrational, and explain your reasoning.

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10 [Intermediate] • Target: Number sets and properties

Explain why the sum of a rational number and an irrational number is always irrational.



11 [Intermediate] • Target: Sequences and series

Find the 15th term of the arithmetic sequence with first term 6 and common difference 4.

12 [Intermediate] • Target: Sequences and series

Find the sum of the first 8 terms of the geometric sequence $2, 6, 18, 54, \ldots$

13 [Intermediate] • Target: Absolute value and inequalities with integers

Find all integer solutions of $|x - 5| \leq 3$.

14 [Intermediate] • Target: Combining number theory concepts in a multi-part problem

This question has three parts. (a) Find the prime factorization of 84 and 126. (b) Use the prime factorizations to find the GCF and LCM. (c) Verify your answer using the identity GCF × LCM = product of the two numbers.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

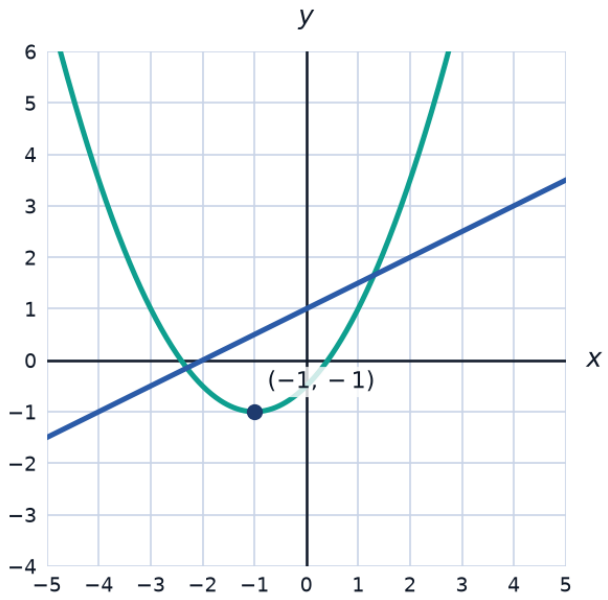
15 [Challenging] • Target: Perfect squares, perfect cubes, and their properties • Cross-Lesson: 2.01 Advanced Algebra & Systems

Find the smallest positive integer n such that $200n$ is a perfect square.

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16 [Challenging] • Target: Perfect squares, perfect cubes, and their properties • Cross-Lesson: 3.01 Advanced Functions

Determine whether 128 is a perfect cube, and justify your answer using prime factorization.



17 [Challenging] • Target: Counting divisors of a number • Cross-Lesson: 4.01 Advanced Trigonometry

Find the number of positive divisors of 360, using its prime factorization $360 = 2^3 \times 3^2 \times 5$.

18 [Challenging] • Target: Working with consecutive integers and integer constraints • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find three consecutive even integers whose sum is 96.

19 [Challenging] • Target: Working with consecutive integers and integer constraints • Cross-Lesson: 6.01 Advanced Coordinate Geometry

The product of two consecutive positive integers is 156. Find the integers.

20 [Challenging] • Target: Digit sum and place value problems • Cross-Lesson: 7.01 Advanced Plane Geometry

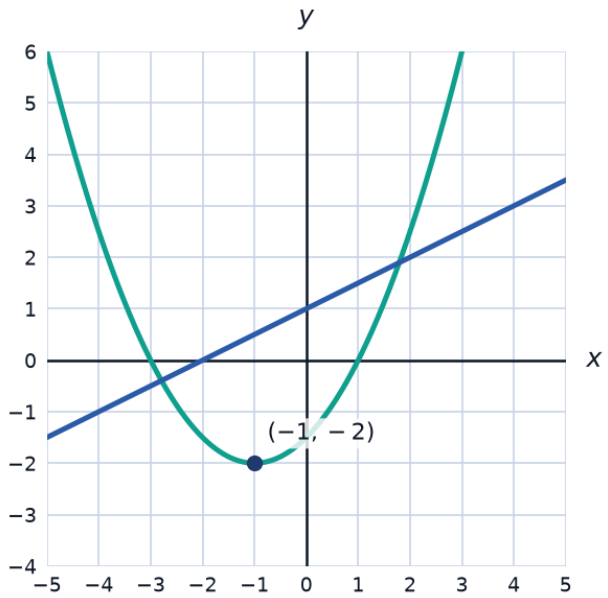
A two-digit number has digit sum 11. When the digits are reversed, the new number is 27 more than the original. Find the original number.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For what value of the digit d is the six-digit integer $452,d76$ divisible by 11?



a. 1

b. 3

c. 5

d. 7

22 [Foundational] • Target: GCF, LCM, and their relationship

Find the remainder when 128 is divided by 9.

a. 1

b. 9

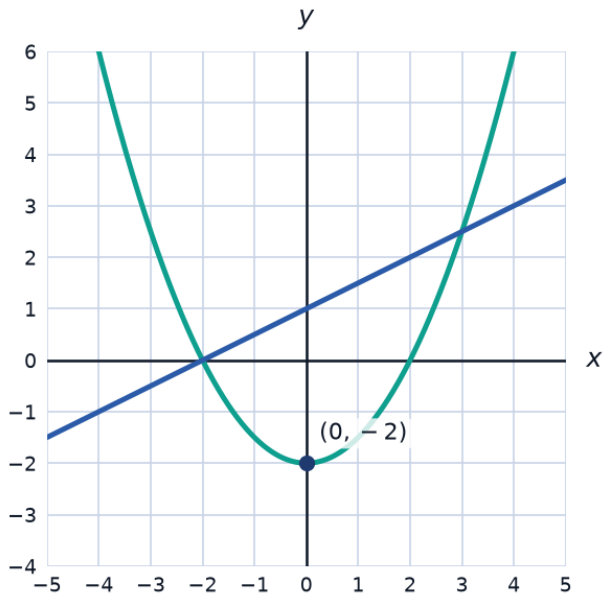
c. 2

d. 3

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23 [Foundational] • Target: Modular arithmetic

Find the remainder when 241 is divided by 12.



a. 0

b. 12

c. 1

d. 2

24 [Foundational] • Target: Sums and products of integer sets

Find the remainder when 78 is divided by 11.

a. 11

b. 0

c. 2

d. 1

25 [Foundational] • Target: Number sets and properties

Find the remainder when 205 is divided by 9.

a. 8

b. 9

c. 7

d. 6

26 [Foundational] • Target: Sequences and series

Find the remainder when 35 is divided by 11.

a. 2

b. 3

c. 1

d. 11

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27 [Foundational] • Target: Absolute value and inequalities with integers

Find the remainder when 78 is divided by 10.

a. 7

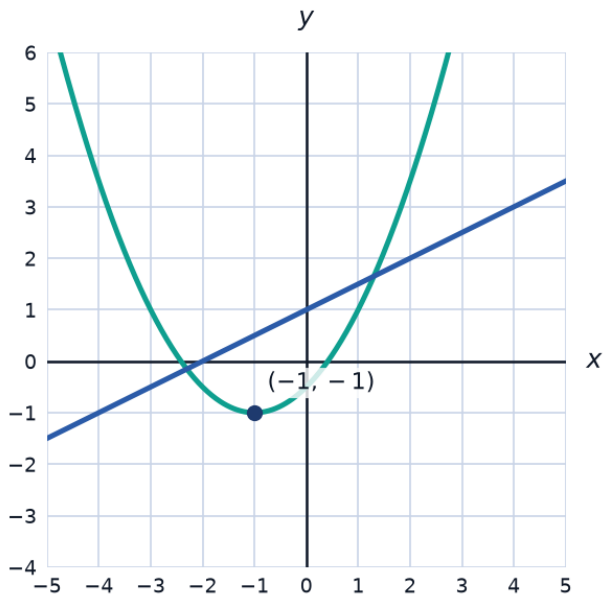
b. 10

c. 9

d. 8

28 [Foundational] • Target: Combining number theory concepts in a multi-part problem

Find the remainder when 74 is divided by 13.



a. 8

b. 9

c. 10

d. 13

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Perfect squares, perfect cubes, and their properties

Find the remainder when 177 is divided by 8.

a. 2

b. 1

c. 8

d. 0

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30 [Intermediate] • Target: Counting divisors of a number

Find the LCM of 54 and 12.

a. 648

b. 6

c. 108

d. 114

31 [Intermediate] • Target: Working with consecutive integers and integer constraints

Find the LCM of 12 and 18.

a. 6

b. 36

c. 216

d. 42

32 [Intermediate] • Target: Digit sum and place value problems

Find the LCM of 24 and 18.

a. 432

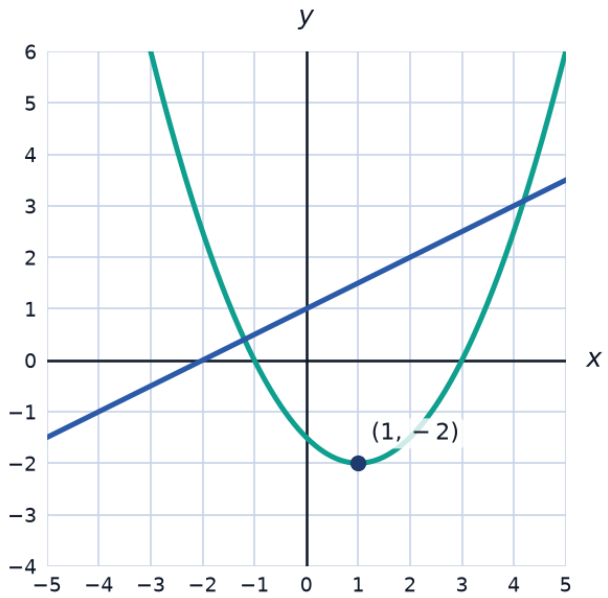
b. 72

c. 78

d. 6

33 [Intermediate] • Target: A multi-part modular arithmetic and remainder problem

Find the LCM of 375 and 135.



a. 50625

b. 3390

c. 15

d. 3375

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the LCM of 100 and 250.

a. 500

b. 25000

c. 50

d. 550

35 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the LCM of 250 and 200.

a. 1050

b. 50000

c. 1000

d. 50

36 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the LCM of 225 and 135.

a. 675

b. 30375

c. 45

d. 720

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the 14th term of the arithmetic sequence with first term -4 and common difference 3.

a. 38

b. 32

c. 37

d. 35

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find the 7th term of the geometric sequence $2, 4, 8, 16, \ldots$

a. 130

b. 125

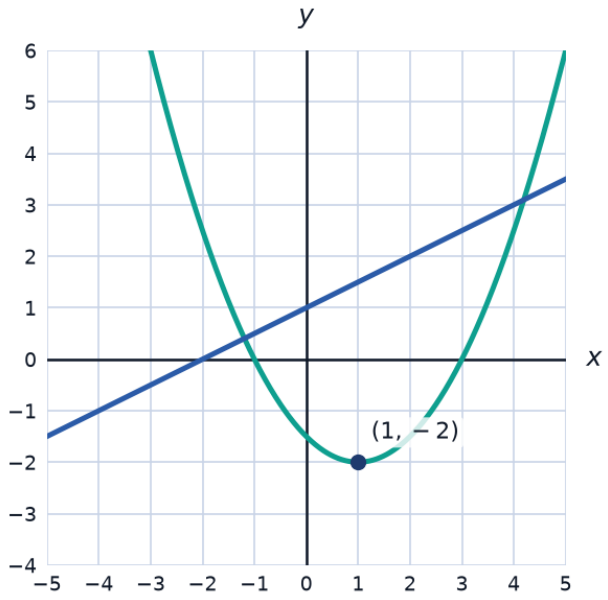
c. 131

d. 128

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Find the 5th term of the geometric sequence $1, 2, 4, 8, \ldots$



a. 19

b. 13

c. 18

d. 16

40 [Challenging] • Target: Divisibility rules and prime factorization • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the number of positive divisors of 528, using its prime factorization $528 = 2^4 \times 3 \times 11$.

a. 18

b. 20

c. 22

d. 4

41 [Challenging] • Target: GCF, LCM, and their relationship • Cross-Lesson: 8.01 Statistics, Counting & Probability

Find the number of positive divisors of 900, using its prime factorization $900 = 2^2 \times 3^2 \times 5^2$.

a. 25

b. 8

c. 29

d. 27

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42 [Challenging] • Target: Modular arithmetic • Cross-Lesson: 2.01 Advanced Algebra & Systems

Find the last digit of 3^{24}, using the pattern of the last digits of powers of 3 (which cycle every 4: 3, 9, 7, 1).

a. 7

b. 3

c. 1

d. 9

43 [Challenging] • Target: Sums and products of integer sets • Cross-Lesson: 3.01 Advanced Functions

Find the last digit of 3^{132}, using the pattern of the last digits of powers of 3 (which cycle every 4: 3, 9, 7, 1).

a. 7

b. 9

c. 1

d. 3

44 [Challenging] • Target: Number sets and properties • Cross-Lesson: 4.01 Advanced Trigonometry

Find the last digit of 4^{129}, using the pattern of the last digits of powers of 4 (which cycle every 2: 4, 6).

a. 4

b. 6

c. 7

d. 9

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Pre-Calculus Concepts

What is the units digit of the integer 7^{2024}?

a. 1

b. 3

c. 7

d. 9

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